

Rajan Vivek

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Education

Stanford University <i>MS Computer Science- Artificial Intelligence (GPA: 4.06 / 4.00)</i> • Courses: NLP, Foundation Models, ML w/ Distribution Shifts, Meta-Learning, Decision Making, Speech Processing	2022-2024 Palo Alto, CA
Georgia Tech <i>BS Electrical Engineering (GPA: 3.97/4.00)</i> • Courses: Statistical ML, Cloud Computing, Python, C++, Signal Processing, Data Structures & Algorithms	2018-2022 Atlanta, GA

Experience

Contextual AI <i>Member of Technical Staff (Research)</i> • Owned real-time hallucination detection feature in company's flagship product end-to-end with 83% F1-score. Required synthetic data generation+eval, reward model fine-tuning, full-stack implementation in production • Led LLM post-training for flagship Grounded LM reaching #1 on leading groundedness benchmark FACTS + best open-weights on customer evals. Built RL infra+reward LM+synthetic data keeping #1 above Gemini 2.5 Pro, GPT-5 • Co-Led training of SoTA reward model (LMUnit), achieving #1 on Reward Bench 2 + FLASK + BigGenBench (as of June 2025) using synthetic data and multi-loss strategy. Led to first-author EMNLP 2025 publication. • 2023 Internship: Extensive LM distillation experiments leading to 2.3X inference speedup via speculative decoding	July 2024-Present Mountain View, CA
Scale AI <i>Machine Learning Research Intern</i> • Performed extensive architecture and data ablations for video/vision language model training, achieving performance surpassing contemporary SoTA models (InstructBLIP, VideoLLaMA, & MPlugOwl) on NextQA video QA	Summer 2023 San Francisco, CA
Stanford NLP Group <i>Researcher</i> • Designed novel representative data selection technique for efficient language model evaluation, allowing rapid discovery of model weaknesses across a large dataset. Led to first author EACL 2024 publication. • Ran many experiments exploring research ideas including benchmark hill-climbing during LM pretraining, data-relatedness metrics, zero-shot vs. fine-tuned performance, meta-learning for synthetic training data generation	Sept. 2022- June 2024 Palo Alto, CA
JP Morgan Chase <i>AI & Data Science Analyst</i> • Trained and productionized a transformer-based entity recognition model for invoice processing. Designed custom data augmentation techniques, improving model performance by 11% (F1-score). AWS Lambda, S3, DynamoDB.	Summer 2022 Jersey City, NJ

Publications

LMUnit: Fine-Grained Evaluation with Natural Language Unit Tests <i>EMNLP Findings 2025</i> • Rajan Vivek* , William Berrios*, Jon Saad-Falcon*, Nandita Naik, Douwe Kiela, Shikib Mehri, et al. (*co-first author) • SoTA reward model (#1 on RewardBench 2 & more) + proposed LLM evaluation paradigm backed by human studies
Anchor Points: Benchmarking Models with Much Fewer Examples <i>EACL Main 2024</i> • Rajan Vivek , Kawin Ethayarajh, Diyi Yang, Douwe Kiela • Novel technique to select data subset for highly efficient eval + elucidate model behavior across the entire dataset
Explainable Activity Recognition for Smart Home Systems <i>ACM Transactions on Interactive Intelligence 2023</i> • Devleena Das, Yasutaka Nishimura, Rajan Vivek , Naoto Takeda, Sean Fish, Thomas Plotx, Sonia Chernova • Framework that applies explainable AI methods (LIME, SHAP) to sensor time series, improving caregiver monitoring

Technical Skills

Foundations: PyTorch, Transformers, Python, distributed training (DDP, FSDP, DeepSpeed ZeRO), experiment tracking
Specialization: RLHF, reward modeling, SFT, DPO, GRPO, synthetic data pipelines, evaluation, agentic systems

Awards

NSF Graduate Research Fellowship Honorable Mention (2022)
Outstanding Project Award, Stanford CS 330: Deep Multi-Task & Meta Learning (2022)
Two 1st place awards, 2nd overall (out of 250 teams), HackGT (2019)